$$\sum_b \phi^*(b)\phi(b) = 1,$$

which makes Alice's density matrix particularly simple:

$$\rho_{a'a} = \psi^*(a)\psi(a'). \tag{7.25}$$

Notice that it only depends on the Alice variables. Perhaps it's not very surprising that everything we need to know about Alice's system is contained in Alice's wave function.

Now, I'm going to prove a key theorem about the eigenvalues of Alice's density matrix, under the assumption of a product state. It is true only for unentangled states and serves to identify them. The theorem says that for any product state, Alice's (or Bob's) density matrix has exactly one nonzero eigenvalue, and that eigenvalue is exactly 1. We begin the theorem by writing the eigenvalue equation for the matrix ρ:

$$\sum_a \rho_{a'a}\alpha_a = \lambda\alpha_{a'}.$$

In other words, the matrix ρ acting on the column vector α gives back the same vector multiplied by an eigenvalue λ. Using the simple form of ρ in Eq. 7.25, we can write

$$\psi(a')\sum_a \psi^*(a)\alpha_a = \lambda\alpha_{a'}. \tag{7.26}$$

Now, you may notice a couple of things. First, the quantity

$$\sum_a \psi^*(a)\alpha_a$$